HEAT WAVE RESPONSE, RECOVERY, FUTURE CHALLENGES, AND THE ROLE OF TECHNOLOGY

A Ground-Level Assessment and Preparedness Study

INDIA SPACE LAB

Project Report

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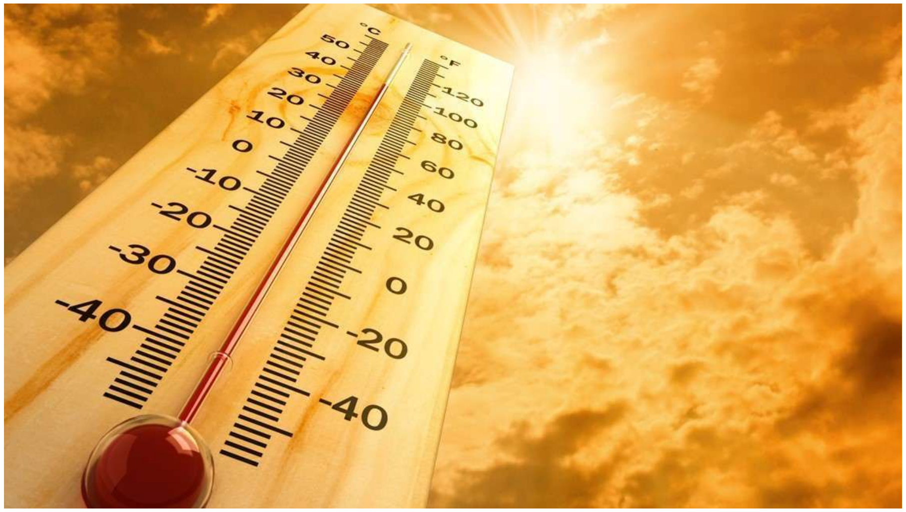
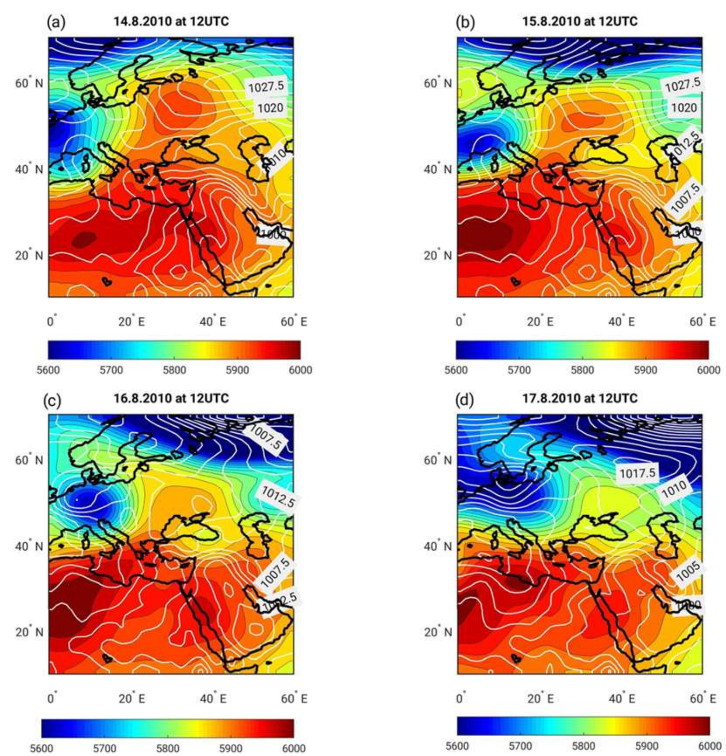
Section 1 — Introduction to Heat Waves

- **Defining the Threat:** A heat wave is a period of abnormally high temperatures exceeding historical seasonal normals, posing extreme risks to human safety and urban systems.
- **Emerging Disaster Status:** Heat waves have escalated from seasonal discomforts to major climate-related disasters in India, driven by global warming, rapid urban sprawl, and deforestation.
- **Cross-Sectoral Impacts:** Multi-system strain affecting public health, water reserves, agricultural yields, informal livelihoods, and energy grids.
- **Focusing on Micro-Climatic Vulnerability:** Sub-Himalayan valleys and mid-altitude urban regions (e.g., Sundernagar) face growing risks due to trapped heat and low baseline thermal inertia.



Section 2 — Science Behind Heat Waves

- **Meteorological Mechanisms:** High-pressure atmospheric systems (synoptic ridging) force air downward, trapping heat near the surface and preventing cloud formation.
- **Urban Heat Island (UHI) Dynamics:** Dense concrete structures, asphalt pavements, and high vehicular density absorb shortwave radiation during the day and re-emit longwave thermal radiation at night.
- **Wet-Bulb Thermodynamics:** Heat stress relies heavily on humidity; when relative humidity is high, human sweat cannot evaporate, halting the body's natural cooling mechanisms ($T_w = f(T_{\text{ambient}}, RH, P)$).
- **Elevation-Dependent Warming (EDW):** Mountain/valley ecosystems experience accelerated warming cycles and thermal inversion trapping.



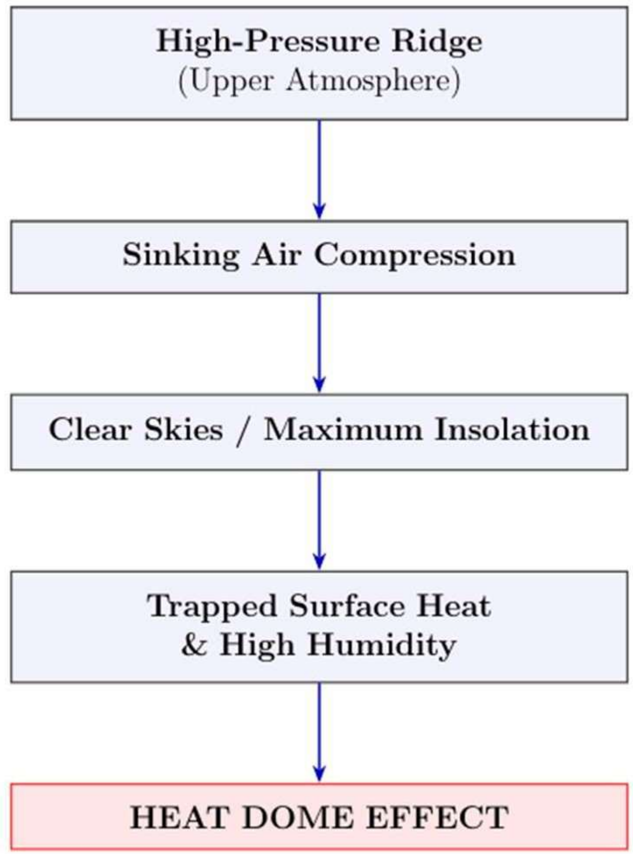
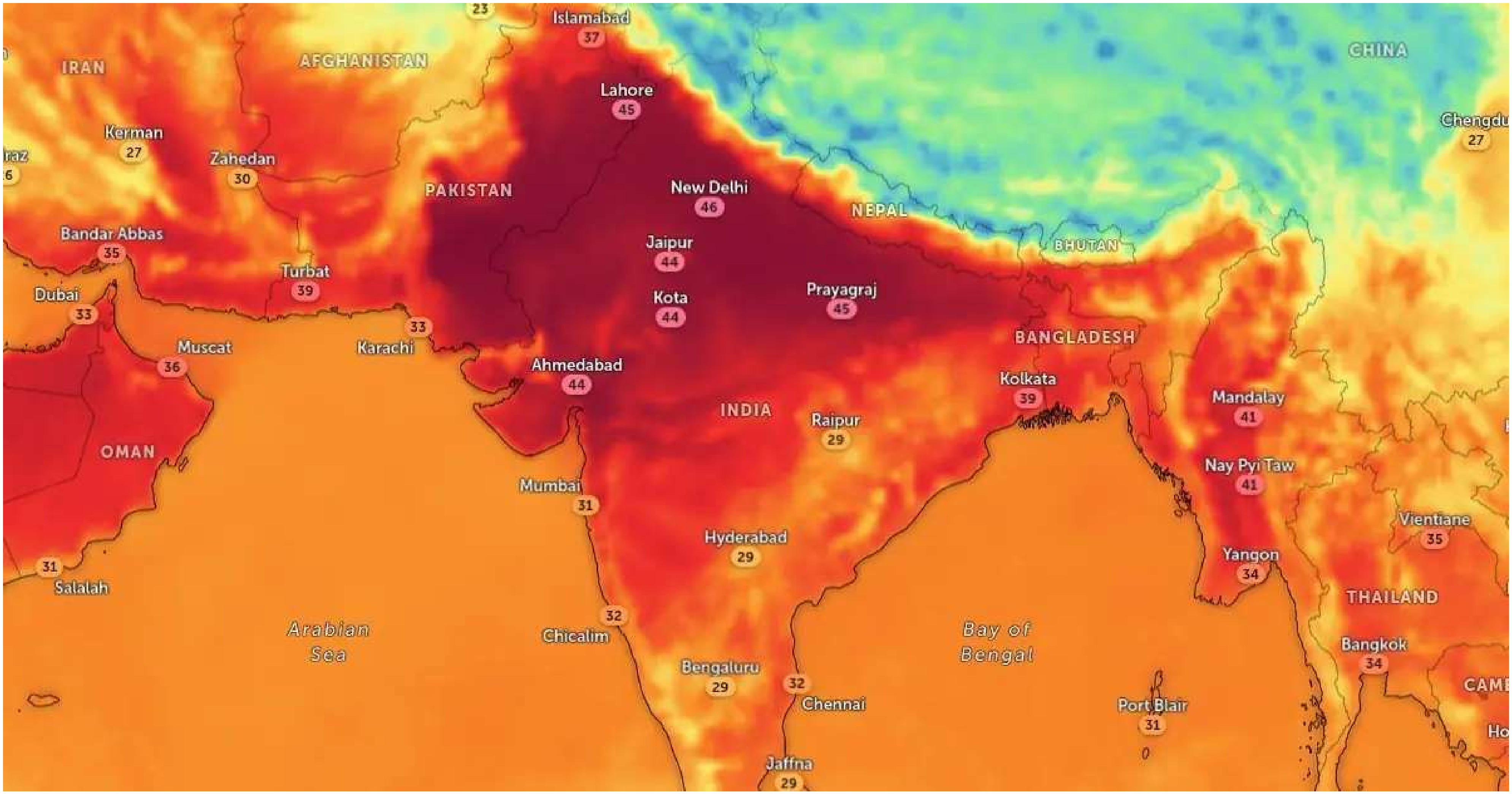


Figure 1: Meteorological Progression of the Atmospheric Heat Dome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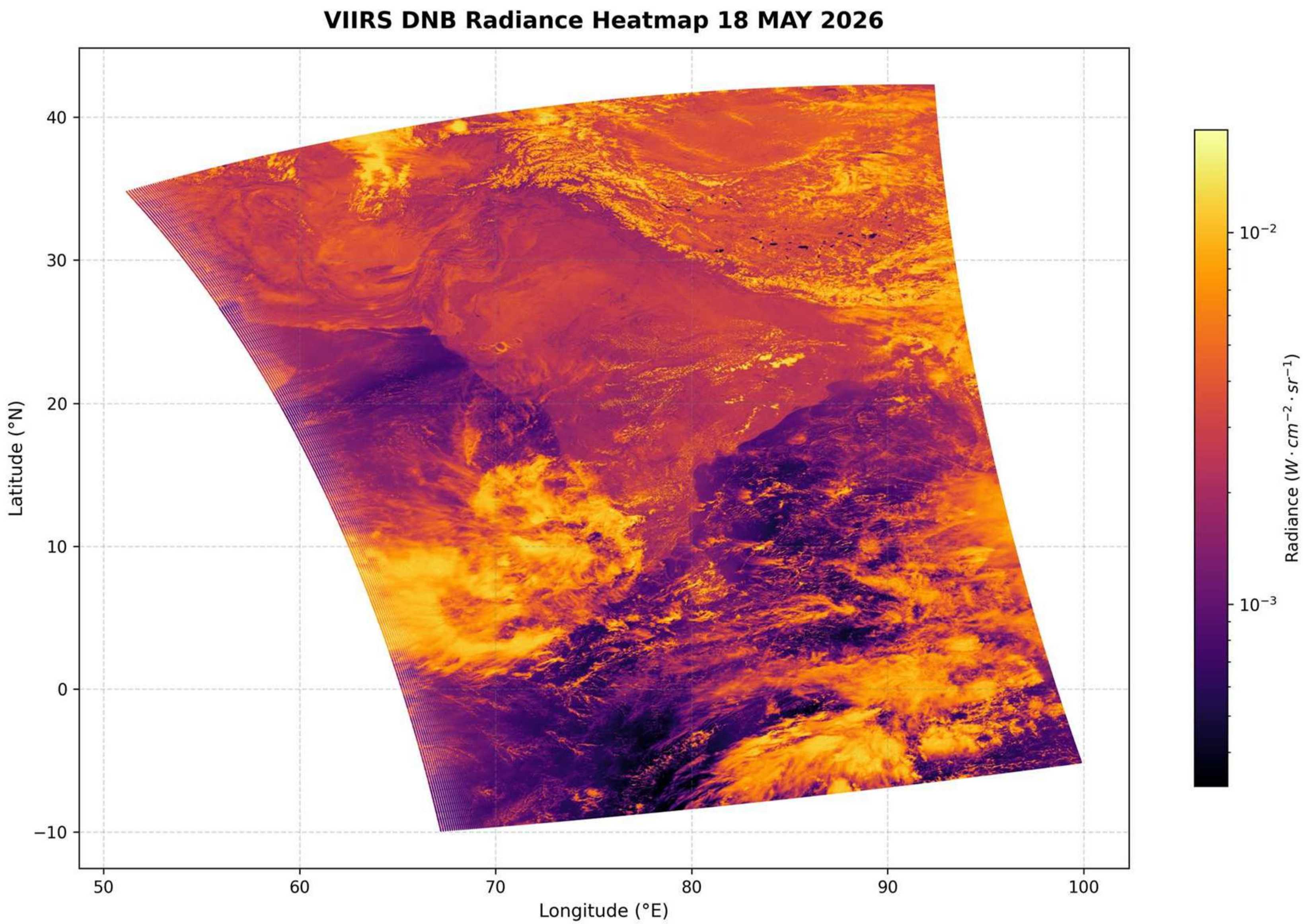
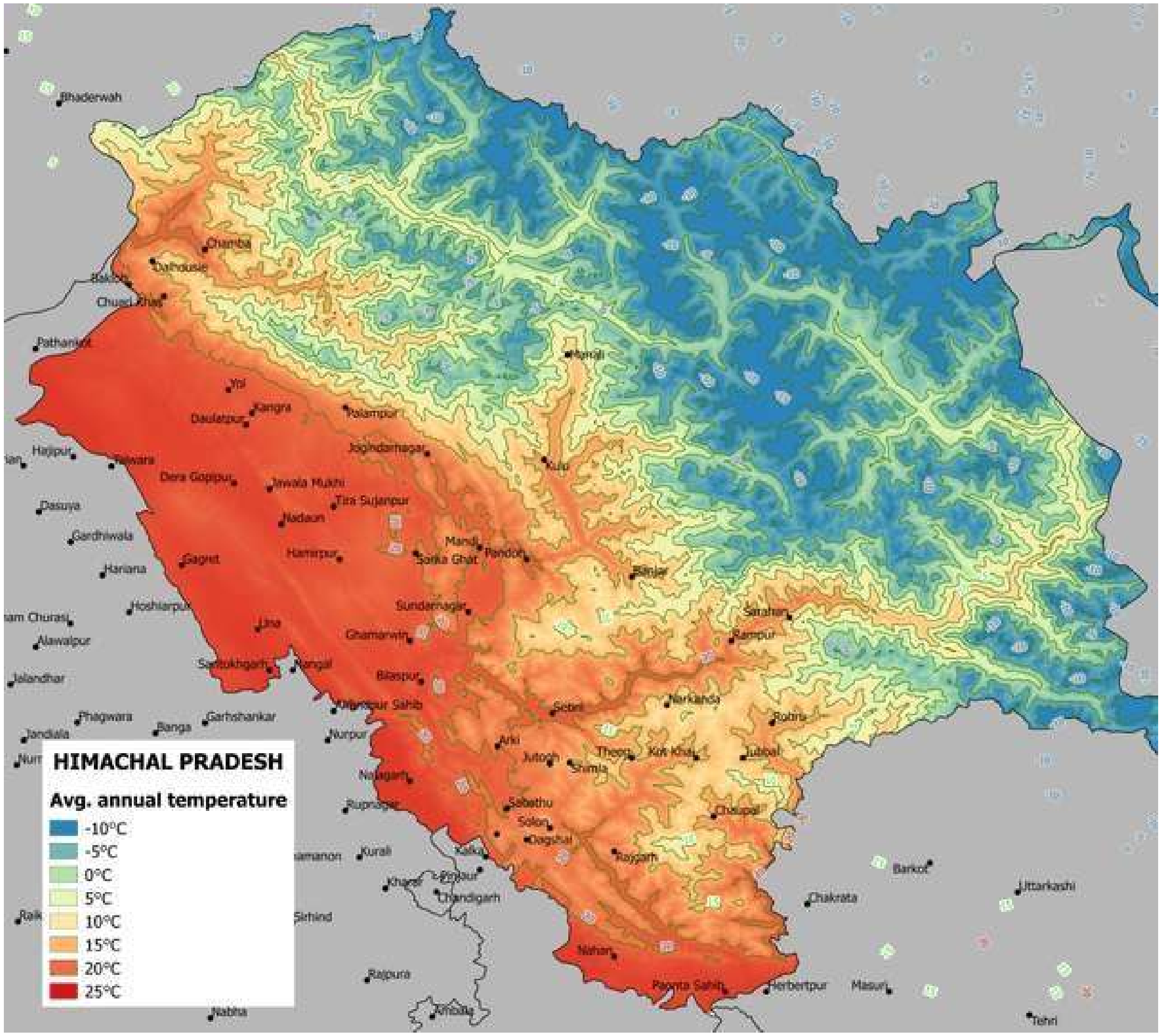

VIIRS DNB Radiance Heatmap 18 MAY 2026
Latitude (°N)
Longitude (°E)
Radiance ($W \cdot cm^{-2} \cdot sr^{-1}$)
10^{-2}
10^{-3}
40
30
20
10
0
−10
50
60
70
80
90
100


HIMACHAL PRADESH
Avg. annual temperature
-10°C
-5°C
0°C
5°C
10°C
15°C
20°C
25°C
Bhaderwah
Chamba
Dalhousie
Pathankot
Manali
Yol
Kangra
Daulatpur
Palampur
Jogindarnagar
Hajipur
Talwara
Dera Gopipur
Jawala Mukhi
Tira Sujanpur
Kulu
Dasuya
Nadaun
Gardhiwala
Hamirpur
Mandi
Pandoh
Gagret
Sarka Ghat
Banjar
Hariana
Hoshiarpur
Sundarnagar
Sarahan
Una
Ghamarwin
Rampur
Alawalpur
Nangal
Jalandhar
Bilaspur
Narkanda
Phagwara
Garhshankar
Sabri
Rohru
Banga
Jandiala
Nurpur
Arki
Jutogh
Theog
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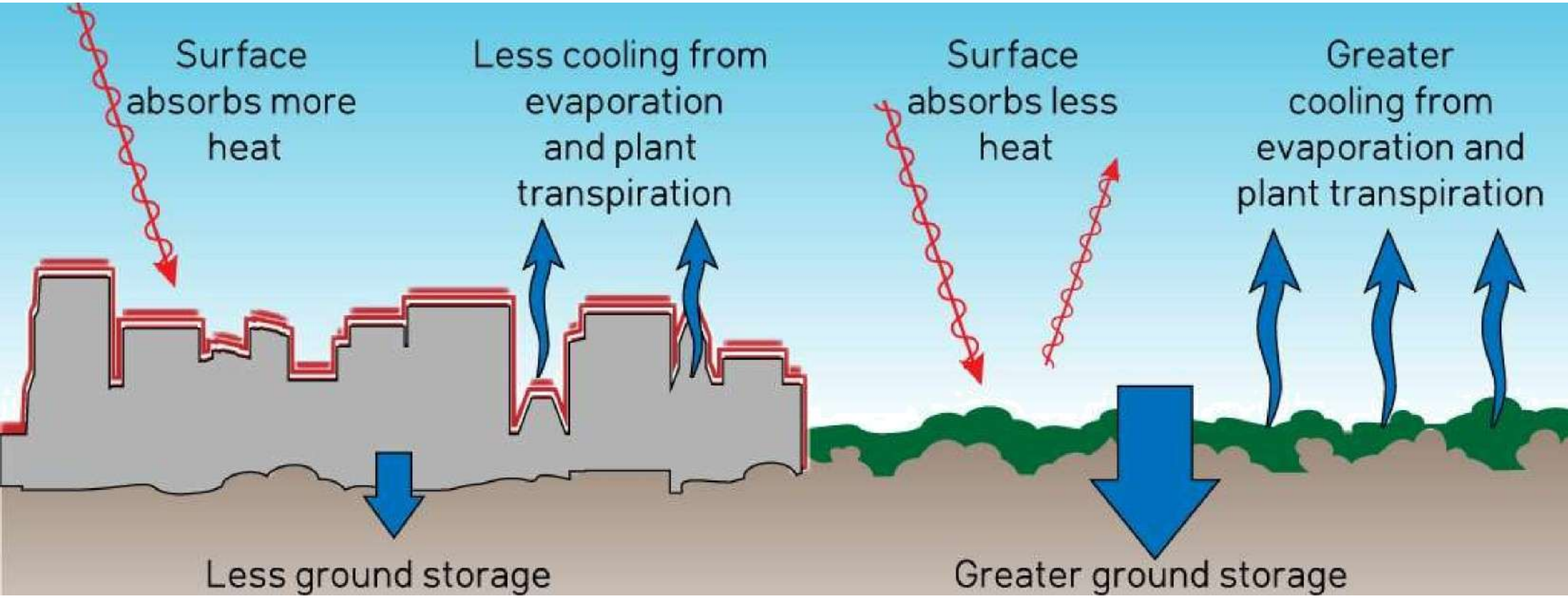
Jubbal
Shimla
Nalagarh
Rupnagar
Sabathu
Chaupal
Solon
Dagshai
Kurali
Kalka
Rajgarh
Barkot
Kharar
Chandigarh
Chakrata
Uttarkashi
Sirhind
Nahan
Rajpura
Paonta Sahib
Herbertpur
Masuri
Nabha
Ambala
Tehri

Urban Heat ”Island E”fiect



of rainfall

of rainfall

Urban

Rural

Section 3 — Environmental Impact and Linked Hazards

- **Hydrological Strain:** Rapid depletion of groundwater tables, drying up of local water bodies, and severe evaporation losses leading to municipal water rationing.
- **Compound Disasters:** Direct links between extreme heat waves, prolonged droughts, elevated ground-level ozone (PM 2.5), and heightened risk of forest/wildfires.
- **Ecological Disruption:** Crop failure, stress on biodiversity, and severe thermal stress on livestock and domestic animals.
- **Energy Grid Overload:** Spikes in peak cooling demand leading to transformer burnout, rolling blackouts, and grid instabilities.

Section 4 — Identifying Vulnerable Groups

- **Occupational High-Exposure Groups:**
- **Traffic Police & Transit Workers:** Prolonged, direct solar radiation and vehicular heat emissions.

- **Street Vendors & Market Laborers:** Lack of permanent shade and dependence on daily informal wages.

- **Construction & Sanitation Workers:** Strenuous physical labor during peak radiation hours (12:00 PM–3:00 PM).

- **Biologically High-Risk Cohorts:**
- **Elderly & Children:** Higher thermal sensitivity, altered thermoregulation, and rapid dehydration rates.

- **Pregnant Women & Chronically Ill:** Heightened cardiovascular/renal vulnerability.

- **Socio-Economically Exposed:** Slum residents and informal settlement communities living under uninsulated tin/asbestos roofs.

Table 1: Demographic Vulnerability Classification for Extreme Heat

Vulnerable Category	Key Demographic Groups	Primary Risk Factors
Occupational Risk	Outdoor workers, construction laborers, street vendors, agricultural workers, traffic police, sanitation workers	Direct solar exposure, heavy physical exertion, limited access to shade or rest breaks.
Physiological Risk	Elderly individuals, infants, young children, pregnant women, individuals with chronic illnesses	Reduced physiological thermoregulation, mobility limitations, dependence on caregivers.
Socioeconomic Risk	Low-income urban households, slum residents, migrant workers	Overcrowded housing, tin or asbestos roofing, lack of ventilation, unreliable water access.

Section 5 — Health Risks to Human Life

- **Direct Thermal Injuries:**
- **Heat Exhaustion:** Dizziness, profuse sweating, rapid pulse, and severe dehydration.
- **Heat Stroke (Medical Emergency):** Core body temperature rising above **40°C**, leading to systemic organ failure and brain tissue damage.
- **Cardiovascular & Renal Stress:** Extreme dilation of blood vessels straining the heart, and severe dehydration leading to Acute Kidney Injury (AKI).
- **Indirect Health Emergencies:** Spikes in vector-borne diseases, food degradation, and disruption of life-saving vaccine cold chains during power outages.

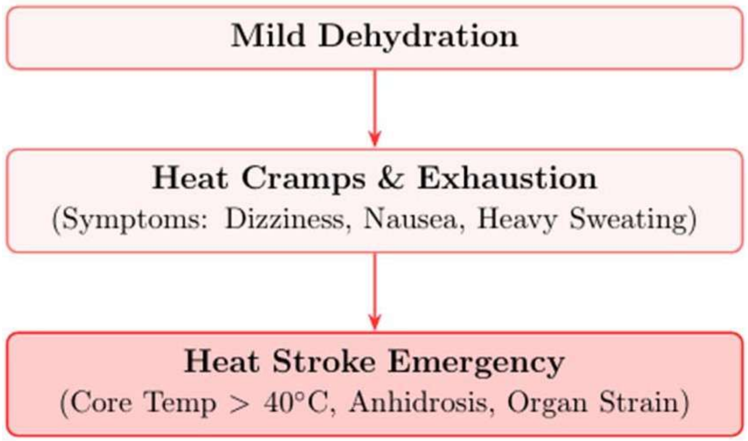


Figure 2: Thermoregulatory Overload Continuum

Section 6

How to Prepare for Heat Waves (Response Strategy)

How to Prepare for Heat Waves (Response Strategy)

- **1. Public Awareness Programs:** Multi-channel educational campaigns on hydration (ORS, electrolytes), shade-seeking behavior, and recognizing early symptoms.

- **2. Situation Reaction Test (SRT) Quizzes:** Deploying short digital/field quizzes to test and measure the actual heat readiness, first-aid knowledge, and behavioral preparedness of common people.

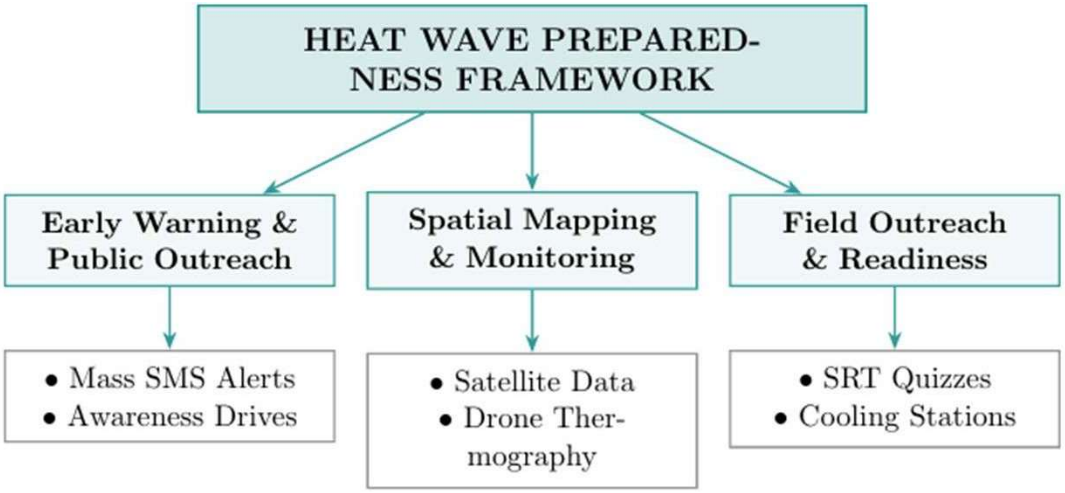
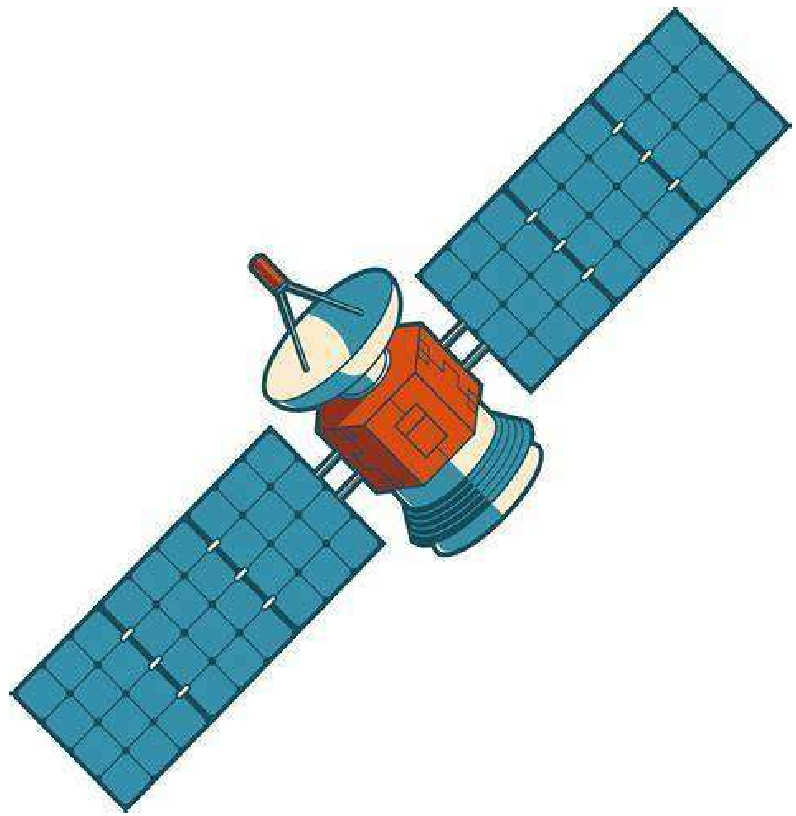


Figure 3: Multidimensional Preparedness Framework

How to Prepare for Heat Waves (Response Strategy)



3. Satellite & Drone Heat Mapping:

- **Satellite Remote Sensing (Landsat/Sentinel):** High-resolution Land Surface Temperature (LST) tracking across wards.
- **UAV/Drone Thermal Surveys:** Micro-level aerial thermal scans to map localized street canyons, high-density market hotspots, and unshaded zones.

How to Prepare for Heat Waves (Response Strategy)

4. Compliant & Planned Construction Activities:
• Updating local building bye-laws for passive ventilation and mandatory reflective roof materials.
• Strategic urban planning for shaded transit stops, solar-powered hydration kiosks, and cooling shelters.

How to Prepare for Heat Waves (Response Strategy)

5. Green Roof & Blue Wall Initiatives:

• **Green Roofs:** Installing vegetative canopy layers on roofs to lower rooftop temperatures (**10°C–18°C** reduction) and reduce indoor heat.

• **Blue Walls:** Wall-mounted vertical cooling panels cooled by sprinkling recycled/greywater, using active evaporative cooling to drop surrounding air temperatures (**2°C–5°C**)

Green Roofs (Solar Radiation Control)	Blue Walls (Active Evaporative Cooling)
• Vegetation layer absorbs solar energy for photosynthesis. • Evapotranspiration cools roof surface below.	• Water mist piped onto exterior wall surfaces. • Evaporation extracts heat, cooling structural mass.

Section 7: Recovery Measures

- **Short-Term Recovery:**
- Restoration of water/power grids using digital complaint portals and 24/7 helplines.
- Immediate medical triage, IV hydration support, and post-event health monitoring for affected populations.

- **Medium-Term Recovery:**
- Retrospectively reviewing SRT quiz data to identify knowledge gaps and refine future Heat Action Plans (HAPs).
- Livelihood support and financial assistance for informal laborers who lost workdays due to extreme weather.

- **Long-Term Infrastructural Retrofits:**
- Scaling up green roof subsidies in low-income/informal settlements.
- Expanding bioclimatic corridors, urban tree afforestation grids, and smart misting infrastructure.

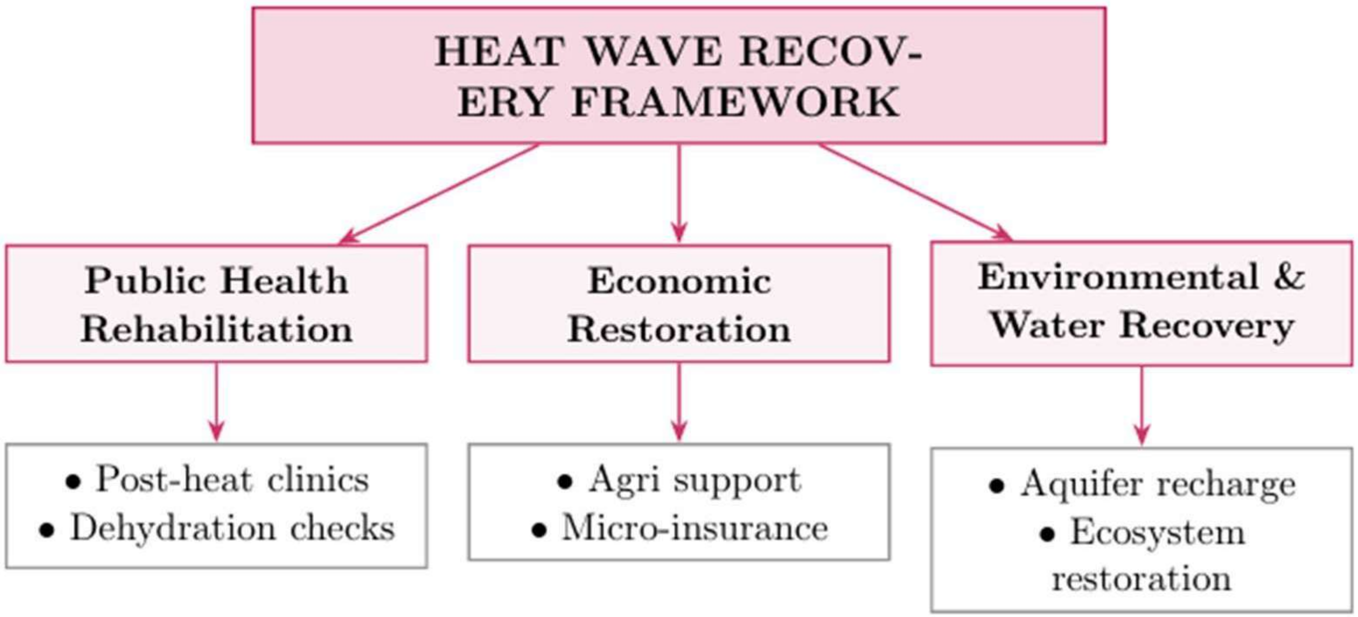


Figure 4: Comprehensive Post-Disaster Recovery Framework

Section 8 — Conclusion & Strategic Vision

- **Proactive over Reactive:** Moving away from temporary emergency relief toward permanent, technology-driven, and bioclimatic urban adaptation.

- **Summary of Integrated Solutions:** Combining satellite/drone spatial analytics, public SRT readiness testing, compliant urban design, and nature-based green/blue infrastructure creates a resilient defense.

- **Call to Action:** Institutionalizing dedicated local Heat Recovery Cells within municipal disaster plans to protect public health and guarantee sustainable urban growth.

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